Differential Equations — Reference

notation

Symbol	Meaning
y', $\dfrac{dy}{dx}$	first derivative of y
y'', $\dfrac{d^2y}{dx^2}$	second derivative
y''', $y^{(4)}$, $y^{(n)}$	third, fourth, n-th derivative
x or t	**independent variable** (the input)
y	**dependent variable** (the unknown function)
$y(x_0)=y_0$	**initial condition**: the value at x_0
c, c_1, c_2	constants from integration
$C=\pm e^c$	a renamed constant: capital C, not c
M_y, $\dfrac{\partial M}{\partial y}$	**partial derivative**: differentiate in y, hold x constant
$M\,dx+N\,dy=0$	**differential form**, the same as $M+Ny'=0$
$F(x,y)=c$	**potential function** of an exact equation
μ ("mu")	**integrating factor**: the multiplier
u, v	a new variable in a substitution
τ ("tau")	half-life
$W(y_1,y_2)$	Wronskian: $y_1y_2'-y_1'y_2$

If the independent variable is time, the book writes t, and y' means $\frac{dy}{dt}$.

terminology

Term	Meaning
differential equation (DE)	an equation with one or more derivatives of an unknown function
ordinary (ODE)	the unknown is a function of one variable
partial (PDE)	the unknown has two or more variables (partial derivatives)
order	the highest derivative in the equation
linear	y and its derivatives appear to the first power only, not multiplied together, not inside other functions
autonomous	$y'=f(y)$: no x on the right side
solution	a function that makes the equation true on an open interval
general solution	the family of all solutions, with constants c
particular solution	one member of the family, with numbers for c
explicit solution	$y=$ a formula in x
implicit solution	an equation in x and y, not solved for y
initial value problem (IVP)	a DE plus initial conditions
interval of validity	the largest open interval that contains x_0 on which the solution works
direction field	short slope marks drawn on a grid
isocline	a curve where the slope $f(x,y)$ is one constant
equilibrium solution	a constant solution $y=k$
separable	$y'=g(x)\,p(y)$: a product of an x part and a y part
lost solution	a solution $y=k$ that a division by $p(y)$ removed
exact	$M\,dx+N\,dy=0$ with $M_y=N_x$
standard form (linear)	$y'+p(x)y=q(x)$: the coefficient of y' is 1
complementary equation	the linear equation with $q=0$: $y'+p(x)y=0$
integrating factor	a multiplier μ that makes an equation exact, or makes the left side $(\mu y)'$
substitution	a new variable that turns a hard equation into a known type
Bernoulli	$y'+p(x)y=q(x)y^n$, $n\neq 0,1$
homogeneous	two meanings: linear with $q=0$ (sections 16, 23), *or* $y'=q(y/x)$ (section 18)
half-life	time for half the amount to decay
steady state	the value Q approaches as $t\to\infty$
terminal velocity	the velocity approached as $t\to\infty$
escape velocity	the smallest launch speed that never falls back
trivial solution	$y=0$
linearly independent	neither function is a constant multiple of the other
fundamental set	two independent solutions. $c_1y_1+c_2y_2$ gives every solution
Wronskian	$W=y_1y_2'-y_1'y_2$. Nonzero means independent

linear or not? — the checklist

A linear equation can be written as

$$a_n(x)y^{(n)}+\cdots+a_1(x)y'+a_0(x)y=g(x)$$

Allowed (still linear)	Not allowed (nonlinear)
coefficients in x: x^2y'', e^xy'	a power of y: y^2, $(y')^3$, $\sqrt{y}$
any $g(x)$ on the right: $\sin x$	a product: yy', yy''
a constant coefficient: $3y'$	y inside a function: $\sin y$, e^y

calculus tables

Derivatives (a, n constants):

Function	Derivative	Note
x^n	nx^{n-1}	power rule
e^{ax}	ae^{ax}	chain rule
$\ln\lvert x\rvert$	$\dfrac{1}{x}$	
$\sin(ax)$	$a\cos(ax)$	
$\cos(ax)$	$-a\sin(ax)$	
$\tan x$	$\sec^2 x$	
$\arctan x$	$\dfrac{1}{1+x^2}$	
fg	$f'g+fg'$	product rule
$\dfrac{f}{g}$	$\dfrac{f'g-fg'}{g^2}$	quotient rule
$f(g(x))$	$f'(g(x))\,g'(x)$	chain rule
$H(y(x))$	$H'(y)y'$	chain rule, y inside
μy	$\mu y'+\mu' y$	product rule, used in 16

Implicit differentiation: differentiate both sides in x. Each y term gets a factor y': $\frac{d}{dx}(y^2)=2yy'$.
Partial derivatives: differentiate in one letter. The other letter is a constant, term by term.

Quick example	Work → answer
$\frac{\partial}{\partial y}(x^2y+5x)$	x^2 is a constant: x^2. $5x$ has no y: 0
$\frac{\partial}{\partial x}(x^2y+3y^2)$	$2xy$. $3y^2$ has no x: 0
$\frac{\partial}{\partial y}(ye^x)$	a plain y gives 1: e^x

Integrals (add $+c$):

Function	Antiderivative
x^n, $n\neq -1$	$\dfrac{x^{n+1}}{n+1}$ divide by $n+1$
$u^{-1/2}$	$2u^{1/2}$ *($\div\frac{1}{2}$ is $\times 2$)*
$\dfrac{1}{x}$	$\ln\lvert x\rvert$
$\dfrac{1}{ax+b}$	$\dfrac{1}{a}\ln\lvert ax+b\rvert$
e^{ax}	$\dfrac{e^{ax}}{a}$ *(the same a below)*
$\sin(ax)$	$-\dfrac{\cos(ax)}{a}$
$\cos(ax)$	$\dfrac{\sin(ax)}{a}$
$\sec^2 x$	$\tan x$
$\dfrac{1}{1+x^2}$	$\arctan x$
$\dfrac{1}{y(a-y)}$	$\dfrac{1}{a}\ln\left\lvert\dfrac{y}{a-y}\right\rvert$ *(partial fractions)*

Integration by parts: $\int u\,dv=uv-\int v\,du$. Pick u = the part that gets simpler when you differentiate (x, x^2, $\ln x$).
u-substitution: $\int f(g(x))g'(x)\,dx=\int f(u)\,du$ with $u=g(x)$. Solve $du=2x\,dx$ by dividing: $x\,dx=\frac{du}{2}$.
Partial fractions: $\dfrac{1}{y(a-y)}=\dfrac{1}{a}\left(\dfrac{1}{y}+\dfrac{1}{a-y}\right)$. Check by adding the two fractions back.

exponent and log rules

Rule	Watch for
$e^{a+b}=e^a e^b$	only a sum in the exponent splits
e^{ab}	does not split: e^{xy} is not $e^x e^y$
$(e^a)^n=e^{an}$	$(e^x)^3=e^{3x}\neq e^{x^3}$
$e^{x^2/2}$	already finished. The 2 never comes down
$\ln(e^{\square})=\square$, $e^{\ln\square}=\square$	ln and e undo each other
$\ln(ab)=\ln a+\ln b$	$\ln(a+b)$ never splits
$k\ln\lvert x\rvert=\ln\lvert x\rvert^k$	so $e^{2\ln x}=x^2$, $e^{-\ln x}=\frac{1}{x}$
$a^t=e^{(\ln a)t}$	any base as base e
$e^{(\cdot)}$ is never 0	divide it out of an equation

Self-check: plug a number into the line before and the line after. They must match.

Free y: undo what is wrapped around it, outside in

Wrapped	Undo with	Watch for
$\ln\lvert y\rvert$	$e^{(\cdot)}$ of the whole side	$e^{a+c}=e^a e^c$. Clearing $\lvert\lvert$ gives $\pm$. Rename $\pm e^c\to C$
y^2	square root	$\pm$ appears; the condition picks the sign (C cannot absorb it)
e^{ky}	take ln	isolate $e^{(\cdots)}$ first
$\frac{1}{2}(\cdots)$	multiply by 2	the 2 hits every term, c too
$\arctan y$	tan	$y=\tan(x+c)$, c stays inside

formula sheet

1–3. Classification

order	highest derivative
linear form	$a_n(x)y^{(n)}+\cdots+a_0(x)y=g(x)$
normal form, first order	$y'=f(x,y)$

4–5. Solutions

verify	substitute y, y', … . Both sides must match
$ay''+by'+cy=0$, try e^{rx}	$ar^2+br+c=0$
$x^2y''+bxy'+cy=0$, try x^r	$r(r-1)+br+c=0$

6–8. General solutions and IVPs

$y'=f(x)$	$y=\int f(x)\,dx+c$
$y^{(n)}=f(x)$	integrate n times, n constants
order n IVP	needs n conditions: $y(x_0),\ldots,y^{(n-1)}(x_0)$

9. Models

growth or decay (Malthus)	$P'=aP$, $P=P_0e^{at}$
logistic (Verhulst)	$P'=aP(1-\alpha P)$, $P\to\frac{1}{\alpha}$
Newton's law of cooling	$T'=-k(T-T_m)$
cooling solution	$T=T_m+(T_0-T_m)e^{-kt}$
glucose with injection	$G'=-\lambda G+r$
epidemic	$I'=rI(S-I)$
free fall	$y''=-g$, $y=-\frac{g}{2}t^2+v_0t+y_0$

10–12. Direction fields

slope at (x_0,y_0)	$m=f(x_0,y_0)$
isocline of slope c	$f(x,y)=c$
equilibrium of $y'=f(y)$	$f(k)=0\Rightarrow y=k$
stable / unstable / semistable	arrows toward / away / mixed

13–14. Separable and autonomous

separable	$y'=g(x)\,p(y)$
separated	$\int\frac{dy}{p(y)}=\int g(x)\,dx+c$
lost solutions	$y=k$ where $p(k)=0$
autonomous concavity	$y''=f'(y)\,f(y)$

15. Exact

test	$M_y=N_x$
potential	$F=\int M\,dx+g(y)$, $F_y=N$
solution	$F(x,y)=c$

16–17. Linear and integrating factors

standard form	$y'+p(x)\,y=q(x)$
integrating factor	$\mu=e^{\int p\,dx}$ (no $+c$)
solution	$\mu y=\int\mu q\,dx+c$
$\mu(x)$ for $M\,dx+N\,dy$	$\frac{M_y-N_x}{N}$ has only x: $\mu=e^{\int\text{it}\,dx}$
$\mu(y)$ for $M\,dx+N\,dy$	$\frac{N_x-M_y}{M}$ has only y: $\mu=e^{\int\text{it}\,dy}$

18–19. Substitutions

$y'=q(y/x)$	$y=ux$, $y'=u+xu'$
$y'=f(ax+by)$	$v=ax+by$, $v'=a+by'$
Bernoulli $y'+py=qy^n$	$v=y^{1-n}$
Bernoulli, linear in v	$v'+(1-n)pv=(1-n)q$

20. Growth and decay

exponential	$Q=Q_0e^{a(t-t_0)}$
half-life	$\tau=\frac{\ln 2}{k}$, $Q=Q_0(\frac{1}{2})^{t/\tau}$
compound, continuous	$Q_0(1+\frac{r}{n})^{nt}$, Q_0e^{rt}
made at s, decays at kQ	$Q\to\frac{s}{k}$
savings with deposits d	$Q_0e^{rt}+\frac{d}{r}(e^{rt}-1)$

21. Cooling and mixing

cooling	$T=T_m+(T_0-T_m)e^{-kt}$
mixing	$Q'=c_{in}r_{in}-\frac{Q}{V(t)}r_{out}$
volume	$V=V_0+(r_{in}-r_{out})t$

22. Mechanics

linear drag	$mv'=-mg-kv$, $v_{term}=-\frac{mg}{k}$
square drag, falling	$mv'=-mg+kv^2$, $v_{term}=-\sqrt{\frac{mg}{k}}$
escape velocity	$v_e=\sqrt{\frac{2gR^2}{h+R}}$

23. Second order homogeneous

standard form	$y''+py'+qy=0$
general solution	$y=c_1y_1+c_2y_2$, $W\neq 0$
Wronskian	$W=y_1y_2'-y_1'y_2$
Abel	$W=Ke^{-\int p\,dx}$
reduction of order	$y_2=uy_1$, $u'=\frac{e^{-\int p}}{y_1^2}$

which method? — first order sorter

Test in this order. Stop at the first yes. One equation can fit more than one row. Any method that fits works.

If the equation …	Use	
has only x on the right: $y'=f(x)$	integrate	(6)
factors as (only x)(only y)	separate	(13)
has no x: $y'=f(y)$	separate, phase line	(14)
fits $y'+p(x)\,y=q(x)$	$\mu=e^{\int p}$	(16)
is $M\,dx+N\,dy=0$ with $M_y=N_x$	potential F	(15)
has $M_y\neq N_x$, but a one-letter ratio	$\mu(x)$ or $\mu(y)$	(17)
fits $y'+py=qy^n$	$v=y^{1-n}$	(19)
has a right side in $\frac{y}{x}$ only	$y=ux$	(18)
has a right side in $ax+by$ only	$v=ax+by$	(18)

which tool? — given → find

The number in parentheses is the concept section.

Given → find	Do this	
an equation → ODE or PDE	count the independent variables	(1)
an equation → order	highest derivative	(2)
an equation → linear or not	the checklist	(3)
a function and a DE → solution?	differentiate, substitute	(4)
a DE → values of r	try e^{rx} or x^r	(5)
DEs and functions → match	count constants, then verify	(5)
$y^{(n)} = f(x)$ → general solution	integrate n times	(6)
general solution, conditions → c	substitute, solve	(7)
a solution → interval of validity	remove bad x, keep x_0's piece	(8)
words → a DE	phrase table	(9)
$y' = f(x,y)$ → slope at a point	evaluate f	(10)
a field → its equation	match the features	(10)
slope c → where on the field	isocline $f(x,y) = c$	(11)
$y' = f(y)$ → constant solutions	solve $f(y) = 0$	(12)
$y(0)$ → long-term behavior	phase line	(12)
separable → solution	separate, integrate, free y	(13)
separable → lost solutions	roots of $p(y)$	(13)
$y' = f(y)$ → where curves bend	sign of $f'(y)f(y)$	(14)
$M\,dx + N\,dy$ → exact?	$M_y = N_x$	(15)
exact → solution	build F, then $F = c$	(15)
$y' + py = q$ → solution	μ, then $\mu y = \int \mu q$	(16)
not exact → multiplier	$\mu(x)$ or $\mu(y)$ test	(17)
$y' = q(y/x)$ or $f(ax+by)$ → solution	substitute	(18)
$y' + py = qy^n$ → solution	$v = y^{1-n}$, linear	(19)
half-life → k, amount → time	$k = \frac{\ln 2}{\tau}$, $t = \frac{\tau \ln(Q_0/Q)}{\ln 2}$	(20)
interest → value or deposit	$Q_0 e^{rt}$, Qe^{-rt}	(20)
one temperature reading → $T(t)$	ratio to the power t/t_1	(21)
tank → $Q(t)$	in − out, linear	(21)
mass, drag → $v(t)$, terminal v	$mv' = F$, linear	(22)
altitude h → escape speed	$v_e = \sqrt{2gR^2/(h+R)}$	(22)
two solutions → fundamental set?	$W \neq 0$	(23)
one solution → second solution	reduction of order	(23)
y_1, y_2 → the equation	$p = -\frac{W'}{W}$	(23)

check before you circle an answer

- Divide, do not multiply, when you separate or solve $du = 2x\,dx$ for $x\,dx$.
- Take $e^{(\cdot)}$ of the whole side, not piece by piece.
- Keep $\pm$ when you clear $|\,|$ or take a square root.
- Power rule: divide by $n+1$. $\int u^{-1/2}\,du = 2u^{1/2}$.
- Keep every negative sign in an exponent.
- Keep one variable name. Do not switch from t to x mid-problem.
- Compare each line with the line before it: copy slips ($t^2 \to t$, a coefficient).
- Lowercase c comes from integration. Capital $C = \pm e^c$ is renamed.
- Check the answer: differentiate it and put it back in the DE.

Differential Equations — Concepts

How to read a worked example. Every example is a puzzle: colored pieces go into the slots of the same color.

Step	What it does
1. PIECES	Read each value off the problem. Give each one a color. Prep (standard form, rearranging) is named in this title.
2. TEMPLATE	The formula, with an empty slot [like this] for each piece, in the piece's color.
3. PLACE	Put each piece in its slot. Then do the arithmetic.
middle steps	One operation each, named by the operation: SOLVE, FREE y, CONDITION.
last: CHECK	Put the answer back into the DE, the condition, or a known fact.

Colors. In sections 1–12, step 1 of each example sets the colors. In 13–19: the x side ($g(x)$, M), the y side ($p(y)$, $f(y)$, N), the condition (x_0, y_0), the multiplier ($p(x)$, μ, f'), the right side ($q(x)$, $q(u)$, n). In 20–22: rate in, rate out or a reading, start values, constants (k, r, m, τ), the long-term value (T_m, steady state). In 23: y_1, y_2, conditions, $p(x)$.

1. what a differential equation is

THE IDEA
A **differential equation** (DE) is a rule about how something changes. $P' = 0.02P$ says "the population grows at 2% of its size per unit time." The answer to a DE is a function, not a number.

Formula	In words
$y(x)$, y', y''	**ordinary** DE (ODE): the unknown has one input variable
$u(x,t)$, $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial t}$	**partial** DE (PDE): the unknown has two or more input variables
$\frac{d\,[\text{dependent}]}{d\,[\text{independent}]}$	the top is the unknown function, the bottom is its input

Quick example	Work → answer
$\frac{dy}{dt} = 3y - t$	one input t: ODE. Dependent y, independent t
$\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2}$	inputs x, t: PDE. Dependent u
$x^2\frac{d^2w}{dx^2} + w = 0$	ODE. Dependent w *(it is the one differentiated)*, independent x

Trap: The letter that is differentiated is the dependent variable, even if it is not y.

2. order

THE IDEA
The **order** is the order of the highest derivative. Count the primes on each derivative. Ignore the powers.

Formula	In words
order = the most primes on one y	(order) = (highest derivative)
$(y')^3$	a first derivative, cubed: order 1
y'''	a third derivative: order 3

Trap: A power is not an order. $y'' + (y')^5 = 0$ is order 2: the highest derivative wins.

Worked: order of $y'' - x(y')^3 = \sin x$

1. PIECES (EVERY DERIVATIVE)
y'': two primes $(y')^3$: one prime, then a power

↓

2. TEMPLATE
order = the larger of [primes] and [primes]

↓

3. PLACE
order = the larger of 2 and 1 = 2

↓

4. CHECK
the cube on y' is a power, not a prime count ✓

Quick example	Work → answer
$y' + 3y = e^x$	highest y': order 1
$(y')^4 + y = 0$	only y', raised to a power: order 1
$\frac{d^3y}{dx^3} + \left(\frac{d^2y}{dx^2}\right)^2 = x$	highest $\frac{d^3y}{dx^3}$: order 3

3. linear or nonlinear

THE IDEA
In a **linear** DE, y and its derivatives are "plain": first power, never multiplied together, never inside a function. The x parts can be anything.

$$[a_n(x)]y^{(n)} + \cdots + [a_1(x)]y' + [a_0(x)]y = [g(x)]$$

Test: look only at the terms with y or a derivative of y. A power, a product of two y terms, or y inside a function makes it **nonlinear**.
Trap: The x parts never decide. x^2y'', e^xy', $\sin(x)\,y$ are all linear terms.

Worked: is $x^2y'' + xy' + y = \cos x$ linear?

1. PIECES
coefficients x^2, x, 1 right side $\cos x$

↓

2. TEMPLATE
$[a_2(x)]y'' + [a_1(x)]y' + [a_0(x)]y = [g(x)]$

↓

3. PLACE
$x^2y'' + xy' + 1\,y = \cos x$ fits the template ⇒ linear

↓

4. CHECK (RED FLAGS)
no y^2, no yy', no $\sin y$ ✓

Quick example	Work → answer
$y' + y^2 = x$	y^2 is a power of y: nonlinear
$yy' = 3$	a product of two y terms: nonlinear
$y'' + \sin y = 0$	y inside sin: nonlinear
$e^xy''' - 4y = x^3$	e^x is a coefficient: linear
$\frac{dy}{dx} = x + y$	move y left: $y' - y = x$: linear

4. solutions, and how to verify one

THE IDEA
A **solution** is a function that makes the DE true for every x in an open interval. To check one, make the pieces y, y', y'' and drop each into its slot in the DE.

1. Pieces: find each derivative the DE needs.
2. Template: the DE with an empty slot for each of y, y', y''.
3. Place, then simplify the left side.
4. Compare with the right side. They must be equal for all x in an interval.

Trap: Equal at a few points only is not a solution. $y = x^2$ makes $y' = 2y$ true at $x = 0$ and $x = 1$ only.
Trap: Constants c stay as letters. The check must work for every value of c.

Which way? — verify or find

Given → find	Do this
a function and a DE → yes or no	differentiate the function and substitute
a DE only → the function	solve: integrate (6), or methods 13–23

Worked: verify $y=\frac{x^2}{3}+\frac{1}{x}$ solves $xy'+y=x^2$

1. PIECES
$y=\dfrac{x^2}{3}+\dfrac{1}{x}$ $\qquad y'=\dfrac{2x}{3}-\dfrac{1}{x^2}$ ($\frac{1}{x}=x^{-1}$, *derivative* $-x^{-2}$)

↓

2. TEMPLATE
$x\cdot\boxed{y'}+\boxed{y}=x^2$

↓

3. PLACE
$x\left(\dfrac{2x}{3}-\dfrac{1}{x^2}\right)+\dfrac{x^2}{3}+\dfrac{1}{x}=\dfrac{2x^2}{3}-\dfrac{1}{x}+\dfrac{x^2}{3}+\dfrac{1}{x}$

↓

4. SIMPLIFY
x^2 terms: $\frac{2}{3}+\frac{1}{3}=1$. $\frac{1}{x}$ terms cancel. Left side $=x^2$

↓

5. CHECK
left $=x^2=$ right ✓ $\qquad y$ breaks at $x=0$: a solution on $(0,\infty)$ and on $(-\infty,0)$

Worked: (second order) verify $y=c_1e^x+c_2e^{-2x}$ solves $y''+y'-2y=0$

1. PIECES
$y=c_1e^x+c_2e^{-2x}$ $\quad y'=c_1e^x-2c_2e^{-2x}$ $\quad y''=c_1e^x+4c_2e^{-2x}$

↓

2. TEMPLATE
$\boxed{y''}+\boxed{y'}-2\boxed{y}=0$

↓

3. PLACE
$(c_1e^x+4c_2e^{-2x})+(c_1e^x-2c_2e^{-2x})-2(c_1e^x+c_2e^{-2x})$

↓

4. COLLECT
e^x: $c_1+c_1-2c_1=0$ $\qquad e^{-2x}$: $4c_2-2c_2-2c_2=0$

↓

5. CHECK
left $=0=$ right, for every c_1, c_2 ✓

Quick example	Work → answer
$y=ce^{2x}$ in $y'=2y$	$y'=2ce^{2x}$, $2y=2ce^{2x}$: yes, every c
$y=x^2$ in $y'=2y$	$2x$ vs $2x^2$: equal at 0, 1 only: no

5. finding r, and matching equations to solutions

THE IDEA
The problem gives a shape, like $y=e^{rx}$ or $y=x^r$, with r unknown. Make the pieces with r as a letter, place them, and factor out the shape. It is never 0, so what is left must be 0.

Formula	In words
$ay''+by'+cy=0$, $\quad y=e^{rx}$	$ar^2+br+c=0$ *(each prime becomes one r)*
$x^2y''+bxy'+cy=0$, $\quad y=x^r$	$r(r-1)+br+c=0$

Worked: find r so that $y=e^{rx}$ solves $y''-y'-6y=0$

1. PIECES
$y=e^{rx}$ $\qquad y'=re^{rx}$ $\qquad y''=r^2e^{rx}$

↓

2. TEMPLATE
$\boxed{y''}-\boxed{y'}-6\boxed{y}=0$

↓

3. PLACE, FACTOR
$r^2e^{rx}-re^{rx}-6e^{rx}=e^{rx}\left(r^2-r-6\right)=0$

↓

4. SOLVE
$e^{rx}\neq0$: $\quad(r-3)(r+2)=0\Rightarrow r=3,\ r=-2$

↓

5. CHECK
$r=3$: $9-3-6=0$ ✓ $\qquad r=-2$: $4+2-6=0$ ✓

Worked: find r so that $y=x^r$ solves $x^2y''-4xy'+6y=0$, $x>0$

1. PIECES
$y=x^r$ $\qquad y'=rx^{r-1}$ $\qquad y''=r(r-1)x^{r-2}$

↓

2. TEMPLATE
$x^2\boxed{y''}-4x\boxed{y'}+6\boxed{y}=0$

↓

3. PLACE, FACTOR
$x^2r(r-1)x^{r-2}-4xrx^{r-1}+6x^r=x^r\left(r(r-1)-4r+6\right)=0$ *(add exponents)*

↓

4. SOLVE
$r^2-5r+6=(r-2)(r-3)=0\Rightarrow r=2,\ r=3$

↓

5. CHECK
$r=2$, $y=x^2$: $x^2(2)-4x(2x)+6x^2=0$ ✓

Matching equations to solutions

1. **Count the constants.** Order n has n constants. One c: first order. c_1, c_2: second order.
2. **Read the form** with the clue table.
3. **Verify the guess** (section 4). A guess is not an answer until it checks.
4. **Cross off the pair.**

The function has	Look for the DE
ce^{kx} only	$y'=ky$
$k+ce^{-ax}$	$y'=a(k-y)$ *(equilibrium at k)*
$c_1e^{r_1x}+c_2e^{r_2x}$	$y''+by'+cy=0$, roots r_1, r_2
$c_1\cos(kx)+c_2\sin(kx)$	$y''+k^2y=0$
$c_1x^{r_1}+c_2x^{r_2}$	$x^2y''+bxy'+cy=0$
a polynomial plus c	$y'=f(x)$: integrate to check

Trap: Same order is not enough. Always finish with the substitution check. A list can have more than one answer per equation: e^{-x} and xe^{-x} both solve $y''+2y'+y=0$.

Worked: match (A) $y'=-3y$, (B) $y''+9y=0$, (C) $y''-y'-2y=0$, (D) $y'=2(4-y)$ to (1) $c_1\cos3x+c_2\sin3x$, (2) ce^{-3x}, (3) $4+ce^{-2x}$, (4) $c_1e^{2x}+c_2e^{-x}$

1. PIECES (COUNT CONSTANTS, READ THE NUMBERS)
two constants: (1) $k=3$, (4) $r=2,-1$ $\qquad$ one constant: (2) $k=-3$, (3) $k=4$, $a=2$

↓

2. TEMPLATE (CLUE TABLE ROWS)
$y''+\boxed{k}^2y=0$ $\quad y'=\boxed{k}y$ $\quad y'=\boxed{a}(\boxed{k}-y)$ $\quad r^2+br+c=0$ at $\boxed{r_1}$, $\boxed{r_2}$

↓

3. PLACE
$y''+9y=0$ is (B) $\quad y'=-3y$ is (A) $\quad y'=2(4-y)$ is (D) $\quad$ (4) goes to (C)

↓

4. CHECK EACH
(B): $y''=-9y\Rightarrow-9y+9y=0$ ✓ (C): $4-2-2=0$, $1+1-2=0$ ✓
(D): $y'=-2ce^{-2x}$, $\ 2(4-4-ce^{-2x})=-2ce^{-2x}$ ✓ A–2, B–1, C–4, D–3

6. general solutions by direct integration

THE IDEA
If the right side has only x, undo the derivative: integrate. Each integration adds one new constant, so order n gives n constants.

Formula	In words
$y'=f(x)\Rightarrow y=\int f(x)\,dx+c$	(function) = (integral of its rate) + (constant)
$y''=f(x)$	integrate twice: c_1, then c_1x+c_2
$\int u\,dv=uv-\int v\,du$	by parts, for xe^{ax}, $x\sin x$, $\ln x$

Trap: Add the constant at each integration, not only at the end. The first c_1 becomes c_1x after the second.

Quick example	Work → answer
$y'=x^3$	$y=\frac{x^4}{4}+c$. Check: $\frac{d}{dx}=x^3$ ✓
$y''=\cos(2x)$	$y'=\frac{\sin2x}{2}+c_1$. $\quad y=-\frac{\cos2x}{4}+c_1x+c_2$

Worked: (by parts) general solution of $y' = xe^{3x}$

1. PIECES
$u = x \to du = dx$ *(simpler when differentiated)* $\quad dv = e^{3x}dx \to v = \frac{e^{3x}}{3}$

↓

2. TEMPLATE
$y = \boxed{u}\boxed{v} - \int \boxed{v}\,\boxed{du}$

↓

3. PLACE
$y = x \cdot \frac{e^{3x}}{3} - \int \frac{e^{3x}}{3}\,dx$

↓

4. FINISH THE INTEGRAL
$\int \frac{e^{3x}}{3}\,dx = \frac{e^{3x}}{9} \quad \Rightarrow \quad y = \frac{xe^{3x}}{3} - \frac{e^{3x}}{9} + c$

↓

5. CHECK
$y' = \frac{e^{3x}}{3} + xe^{3x} - \frac{e^{3x}}{3} = xe^{3x}$ ✓ *(product rule on the first term)*

7. initial value problems

THE IDEA
The general solution is a whole family of curves. An **initial condition** is a point the curve must pass through. It picks one curve: the **particular solution**.

Formula	In words
$\boxed{y_0} = y(\boxed{x_0})$	put the point into the general solution, solve for c
order $n \Rightarrow n$ conditions	$y(x_0), y'(x_0), \ldots, y^{(n-1)}(x_0)$
point $(1,2)$	means $y(1) = 2$

Trap: A y' condition goes into y', not into y. Differentiate the general solution first.

Quick example	Work → answer
$y' = 3x^2, \quad y(2) = 5$	$y = x^3 + c. \quad 5 = 8 + c. \quad y = x^3 - 3$
$y = 3 + ce^{-2x}, \quad y(0) = 7$	$7 = 3 + c. \quad y = 3 + 4e^{-2x}$

Worked: $y'' = 6x, \quad y(0) = 1, \quad y'(0) = -2$

1. PIECES
general: $y' = 3x^2 + c_1, \quad y = x^3 + c_1 x + c_2 \qquad$ conditions: $x_0 = 0, y_0 = 1, y_0' = -2$

↓

2. TEMPLATE (ONE EQUATION PER CONDITION)
$\boxed{y_0'} = 3\boxed{x_0}^2 + c_1 \qquad \boxed{y_0} = \boxed{x_0}^3 + c_1\boxed{x_0} + c_2$

↓

3. PLACE
$-2 = 0 + c_1 \Rightarrow c_1 = -2 \qquad 1 = 0 + 0 + c_2 \Rightarrow c_2 = 1 \qquad y = x^3 - 2x + 1$

↓

4. CHECK
$y(0) = 1$ ✓ $\quad y'(0) = -2$ ✓ $\quad y'' = 6x$ ✓

Worked: two constants: $y = c_1 e^x + c_2 e^{-2x}, \quad y(0) = 3, \quad y'(0) = 0$

1. PIECES
$y' = c_1 e^x - 2c_2 e^{-2x} \qquad x_0 = 0, y_0 = 3, y_0' = 0$

↓

2. TEMPLATE
$c_1 e^{\boxed{x_0}} + c_2 e^{-2\boxed{x_0}} = \boxed{y_0} \qquad c_1 e^{\boxed{x_0}} - 2c_2 e^{-2\boxed{x_0}} = \boxed{y_0'}$

↓

3. PLACE
$e^0 = 1$: $\quad c_1 + c_2 = 3 \qquad c_1 - 2c_2 = 0$

↓

4. SOLVE THE PAIR
$c_1 = 2c_2 \Rightarrow 3c_2 = 3 \Rightarrow c_2 = 1, \quad c_1 = 2 \qquad y = 2e^x + e^{-2x}$

↓

5. CHECK
$y(0) = 2 + 1 = 3$ ✓ $\quad y'(0) = 2 - 2 = 0$ ✓

8. interval of validity, explicit and implicit

THE IDEA
A solution lives on one unbroken stretch of the x axis. Find where it breaks, cut the line there, and keep the piece that holds the start point x_0.

Formula	In words
$\frac{\cdots}{\square}$	breaks where $\square = 0$
$\sqrt{\square}$	needs $\square \geq 0$ (> 0 if it is also in a denominator)
$\ln(\square)$	needs $\square > 0$
interval = piece that holds $\boxed{x_0}$	the largest open interval through the start point

Explicit: y alone, $y = \ldots$ $\quad$ **Implicit**: an equation in x and y, like $x^2 + y^2 = 25$. Check an implicit solution with implicit differentiation.

Worked: $y = \frac{1}{x-2}$ ***solves*** $y' = -y^2, \quad y(3) = 1$. ***Find the interval***

1. PIECES
break: $x = 2$ *(x − 2 = 0)* $\qquad$ start $x_0 = 3$

↓

2. TEMPLATE
cut at $\boxed{\text{break}}$, keep the piece that holds $\boxed{x_0}$

↓

3. PLACE
pieces $(-\infty, 2)$ and $(2, \infty)$. $\quad 3$ is in $(2, \infty)$ *(with $y(0) = -\frac{1}{2}$ instead: $(-\infty, 2)$)*

↓

4. CHECK THE SOLUTION
$y' = -(x-2)^{-2}, \quad -y^2 = -(x-2)^{-2}$ ✓ $\quad y(3) = 1$ ✓

Worked: (implicit) $x^2 + y^2 = 25$ ***and*** $y' = -\frac{x}{y}$, ***through*** $(3,4)$

1. PIECES
curve $x^2 + y^2 = 25 \qquad$ point $(3,4)$, so $y_0 = 4$

↓

2. TEMPLATE (IMPLICIT DIFFERENTIATION)
$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0, \quad$ with $\frac{d}{dx}(y^2) = 2y\boxed{y'}$

↓

3. PLACE
$2x + 2yy' = 0 \Rightarrow y' = -\frac{x}{y}$, the DE ✓

↓

4. EXPLICIT PIECE
$y_0 = 4 > 0 \Rightarrow$ keep $+$: $\quad y = \sqrt{25 - x^2}$

↓

5. INTERVAL, CHECK
$25 - x^2 > 0$ *(y = 0 makes y′ undefined)* $\Rightarrow (-5, 5) \qquad$ holds $x_0 = 3$ ✓

9. models: from words to a DE

THE IDEA
"Rate of change" is a derivative. Translate the sentence phrase by phrase, then match the result to a model with a known solution.

Words	Math
the rate of change of y	$\frac{dy}{dt}$
is proportional to y	$= ky$
is proportional to the square of y	$= ky^2$
is inversely proportional to y	$= \frac{k}{y}$
is proportional to the difference of T and T_m	$= k(T - T_m)$
is proportional to the product of I and $S - I$	$= kI(S - I)$
decreases, decays, is absorbed	a minus sign, with $k > 0$
is added at a constant rate r	$+r$

Formula	In words
$P' = aP \Rightarrow P = \boxed{P_0}e^{\boxed{a}t}$	(amount) = (start)(growth factor)
$a = \frac{\ln(P/P_0)}{t}$	rate from two readings
$T = \boxed{T_m} + (\boxed{T_0} - \boxed{T_m})e^{-\boxed{k}t}$	(temp) = (room) + (start gap)(fade)

Quick example	Work → answer
decays at a rate proportional to the amount A	$\frac{dA}{dt} = -kA,\ k > 0$
rumor: product of heard R and not heard, town of 5000	$\frac{dR}{dt} = kR(5000 - R)$

Worked: growth $P' = 0.02P$, $P(0) = 1000$. Find $P(10)$

1. PIECES
$a = 0.02$ $P_0 = 1000$

↓

2. TEMPLATE
$P = \boxed{P_0}e^{\boxed{a}t}$

↓

3. PLACE
$P = 1000e^{0.02t}$ $P(10) = 1000e^{0.2} = 1221$

↓

4. CHECK
2% a year for 10 years: a bit more than 20% growth ✓

Worked: (reverse) a population doubles in 35 years. Find a

1. PIECES
$P/P_0 = 2$ *(only the ratio matters)* $t = 35$

↓

2. TEMPLATE
$$a = \frac{\ln(\boxed{P/P_0})}{\boxed{t}}$$

↓

3. PLACE
$$a = \frac{\ln 2}{35} = \frac{0.6931}{35} = 0.0198 \text{ per year}$$

↓

4. CHECK
$e^{0.0198(35)} = e^{0.693} = 2$ ✓

Worked: cooling $T' = -0.1(T - 20)$, $T(0) = 90$. Find $T(5)$

1. PIECES
$k = 0.1$ $T_m = 20$ $T_0 = 90$

↓

2. TEMPLATE
$T = \boxed{T_m} + (\boxed{T_0} - \boxed{T_m})e^{-\boxed{k}t}$

↓

3. PLACE
$T = 20 + (90 - 20)e^{-0.1t} = 20 + 70e^{-0.1t}$

↓

4. EVALUATE
$T(5) = 20 + 70e^{-0.5} = 20 + 70(0.6065) = 62.5$

↓

5. CHECK
between 20 and 90 ✓ $t \to \infty$: $T \to 20$, the room ✓

Worked: free fall $y'' = -9.8$, $y(0) = 24.5$ m, $y'(0) = 19.6$ m/s

1. PIECES
$g = 9.8$ $v_0 = 19.6$ $y_0 = 24.5$

↓

2. TEMPLATE
$$y = -\frac{\boxed{g}}{2}t^2 + \boxed{v_0}t + \boxed{y_0}$$

↓

3. PLACE
$$y = -\frac{9.8}{2}t^2 + 19.6t + 24.5 = -4.9t^2 + 19.6t + 24.5 \qquad y' = -9.8t + 19.6$$

↓

4. TOP: $y' = 0$
$t = 2$ s. $y(2) = -19.6 + 39.2 + 24.5 = 44.1$ m

↓

5. GROUND: $y = 0$
$\div(-4.9)$: $t^2 - 4t - 5 = (t - 5)(t + 1) = 0 \Rightarrow t = 5$ s *(-1 is before launch)*

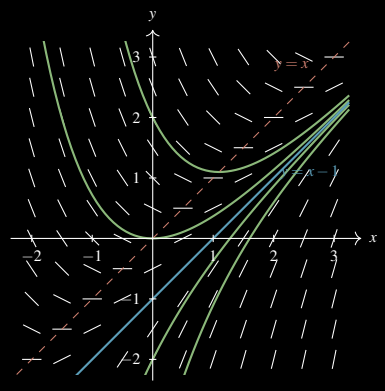
↓

6. CHECK
$y(5) = -122.5 + 98 + 24.5 = 0$ ✓

10. direction fields

THE IDEA
$y' = f(x, y)$ tells you the slope of the solution curve at every point, without solving anything. Draw a short dash with that slope at each grid point. Solution curves follow the dashes, like water follows a current.

Formula	In words
$m = f(\boxed{x_0}, \boxed{y_0})$	(slope at a point) = the right side evaluated there
$f = 0$	flat dashes
$f > 0$ / $f < 0$	dashes go up / down to the right



The field is for $y' = x - y$. Green: solution curves. **Blue**: the solution $y = x - 1$. Dashed red: the slope-0 line $y = x$. Every curve bends toward the blue line.

Quick example	Work → answer
$y' = x - y$ at $(1, 0)$	$1 - 0 = 1$, up at 45°
at $(0, 1)$	$0 - 1 = -1$, down at 45°
at $(2, -1)$	$2 + 1 = 3$, steep up
at $(0, 0)$	0, flat *(on the red line)*

Read a field, match an equation:

Feature of the field	What it tells you
same slope along each horizontal row	no x: autonomous, $y' = f(y)$
same slope along each vertical column	no y: $y' = f(x)$
flat dashes along a curve	that curve is $f(x, y) = 0$
a row of flat dashes at $y = k$	equilibrium $y = k$
slope sign in each region	sign of f there

Which way? — equation or field

Given → find	Do this
equation → field	evaluate f at grid points, draw the dashes
field → equation	test the features above, one at a time
field and a start point → curve	start there, follow the dashes both ways

Worked: which of $y' = y$, $y' = x$, $y' = x - y$ fits the field above?

1. PIECES (FEATURES OF THE FIELD)
flat along $y = x$ rows change columns change

↓

2. TEMPLATE (WHAT A CANDIDATE MUST DO)
flat where $f = 0$ is $\boxed{\text{the flat line}}$ rows $\boxed{\text{change}}$: f has x columns $\boxed{\text{change}}$: f has y

↓

3. PLACE EACH CANDIDATE
$y' = y$: no x, rows would match no $y' = x$: flat on $x = 0$ no
$y' = x - y$: flat on $y = x$, has x and y match

↓

4. CHECK
$f(1, 0) = 1$: the dash at $(1, 0)$ rises at 45° ✓

Worked: (reverse) sketch the solution of $y' = x - y$ through $(0,1)$

1. PIECES
start $(0,1)$ $\quad f = x - y$

↓

2. TEMPLATE
$m = f(\boxed{x_0}, \boxed{y_0})$

↓

3. PLACE
$m = 0 - 1 = -1$: down to the right

↓

4. FOLLOW THE DASHES
falls until it meets $y = x$ (flat there), then rises, bending toward $y = x - 1$

↓

5. CHECK
the exact solution $x - 1 + 2e^{-x}$ passes $(0,1)$ and approaches $x - 1$ ✓

11. isoclines

THE IDEA
An **isocline** is a curve where every dash has the same slope. Draw a few of these first, then fill dashes along each one. It is faster than point by point.

Formula	In words
$f(x,y) = \boxed{c}$	the isocline of slope c
$c = 0$	where curves are flat: their peaks and valleys

Quick example	Work → answer
$y' = x - y$, $c = 0$	$y = x$: flat dashes
$y' = x - y$, $c = 1$	$y = x - 1$: slope 1 dashes on a slope-1 line ⇒ the line is a solution
$y' = x^2 + y^2$, slope 4	$x^2 + y^2 = 4$: circle of radius 2

Worked: isoclines of $y' = x + y$ for $c = 0,\ 1,\ -1$

1. PIECES
$f = x + y$ $\quad$ slopes $c = 0$, $c = 1$, $c = -1$

↓

2. TEMPLATE
$x + y = \boxed{c} \quad \Rightarrow \quad y = \boxed{c} - x$

↓

3. PLACE
$y = -x$: flat dashes $\quad y = 1 - x$: slope 1 $\quad y = -1 - x$: slope -1

↓

4. READ
on $y = -1 - x$ the dashes have slope -1, the same as the line ⇒ the line is a solution

↓

5. CHECK
$y = -1 - x$: $\quad y' = -1$ and $x + y = -1$ ✓

12. equilibrium solutions and long-term behavior

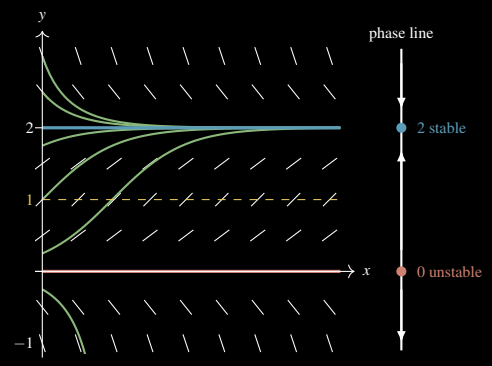
THE IDEA
For $y' = f(y)$, a value k with $f(k) = 0$ means "no change, ever": $y = k$ is an **equilibrium**. Between equilibria, y only goes up or only goes down. Arrows on a vertical line (the **phase line**) show which.

Formula	In words
$f(\boxed{k}) = 0$	$y = k$ is an equilibrium
$f > 0$ on an interval	arrow up: y increases
$f < 0$ on an interval	arrow down: y decreases

Type	Arrows
stable	both point toward it
unstable	both point away
semistable	toward on one side, away on the other

Long term: a solution follows the arrow of its interval to the next equilibrium. If there is none in that direction, it goes to $\pm\infty$.

Trap: Solution curves cannot cross an equilibrium line. A curve that starts between two equilibria stays between them.



The field is for $y' = y(2 - y)$, with its phase line on the right. Dashed yellow: $y = 1$, where the S-curves bend (section 14).

Worked: equilibria, stability, and limits for $y' = y(2 - y)$

1. PIECES (ROOTS OF f)
$y(2 - y) = 0 \Rightarrow y = 0,\ y = 2$ $\quad$ test values -1, 1, 3, one per interval

↓

2. TEMPLATE
sign of $f(\boxed{\text{test}})$ in each interval cut by $\boxed{\text{roots}}$

↓

3. PLACE
$f(-1) = -3$ down $\quad f(1) = 1$ up $\quad f(3) = -3$ down

↓

4. CLASSIFY, LIMITS
2: up from below, down from above ⇒ stable $\quad$ 0: both away ⇒ unstable
$y(0) = 1$ or 3: $y \to 2$ $\quad y(0) = 0$: stays $\quad y(0) = -1$: $y \to -\infty$

↓

5. CHECK
section 14 solves it: $y = \frac{2}{1 + e^{-2x}}$ starts at 1 and goes to 2 ✓

Quick example	Work → answer
$y' = (y - 1)^2$	root 1. $\quad f(0) = 1$ up, $f(2) = 1$ up: **semistable**
build: -1 stable, 3 unstable	$y' = (y + 1)(y - 3)$: $\quad f(0) = -3$ down, $f(\pm 4) > 0$ ✓
$y' = x - y$: constant solution?	$0 = x - k$ for all x fails: none *(not autonomous)*

13. separable equations

THE IDEA
If the right side is a product (only x)×(only y), put every y on the left with dy and every x on the right with dx. Then integrate each side in its own letter.

$$y' = \boxed{g(x)} \cdot \boxed{p(y)} \quad \Rightarrow \quad \int \frac{dy}{\boxed{p(y)}} = \int \boxed{g(x)}\,dx + c$$

Formula	In words
$y' = g(x)\,p(y)$	(rate) = (only x)(only y): separable
$h(y)\,y' = g(x)$, $\quad h = \dfrac{1}{p}$	Trench form. h can be any y function: $\frac{1}{y}$, e^{-3y}
$H(y) = G(x) + c$	$H' = h$, $\quad G' = g$: the implicit solution

Why it works: chain rule. If $H' = h$, then $\frac{d}{dx}H(y) = h(y)\,y'$. So $h(y)\,y' = g(x)$ says $\frac{d}{dx}H(y) = \frac{d}{dx}G(x)$, and $H(y) = G(x) + c$.

Which way? — separable or not

Quick example	Work → answer
$y' = xy + x$	factor: $x(y + 1)$ separable
$y' = e^{x+y}$	a sum in the exponent splits: $e^x e^y$ separable
$y' = y^2 \sin x$	$\sin x \cdot y^2$ separable
$y' = x + y$	a sum that will not factor not separable *(linear: 16)*
$y' = e^{xy}$	a product inside an exponent does not split not separable

Procedure

1. Separate: divide by $p(y)$: $\frac{dy}{p(y)} = g(x)\,dx$.
2. Integrate both sides. Add one $+c$, on the x side, now.
3. Free y: undo what is wrapped around it, outside in (reference table).
4. Lost solutions: test each $y = k$ with $p(k) = 0$.
5. IVP: put in (x_0, y_0) on the easiest line, often the implicit one.

Lost solutions

Formula	In words
$p(\boxed{k}) = 0$	dividing by $p(y)$ assumed $p \neq 0$: each root $y = k$ is a candidate
$y = k$, $y' = 0$ in the original DE	a true statement means $y = k$ is a solution
some C gives $y = k$	write "C any real number, including 0"
no C gives $y = k$	list $y = k$ separately

These are exactly the **equilibria** of section 12.

Worked: $y' = 4x(y+2)$, $y(0) = 1$

1. PIECES
$g = 4x$ $\quad p = y+2$ $\quad (x_0, y_0) = (0,1)$

↓

2. TEMPLATE
$$\int \frac{dy}{\boxed{p(y)}} = \int \boxed{g(x)}\,dx + c$$

↓

3. PLACE, INTEGRATE
$$\int \frac{dy}{y+2} = \int 4x\,dx \quad \Rightarrow \quad \ln|y+2| = 2x^2 + c$$

↓

4. FREE y (e OF THE WHOLE SIDE)
$$|y+2| = e^c e^{2x^2} \quad \Rightarrow \quad y+2 = \pm e^c e^{2x^2} = Ce^{2x^2} \quad \Rightarrow \quad y = -2 + Ce^{2x^2}$$

↓

5. LOST SOLUTION
$p(-2) = 0$: $y = -2$ is lost, but $C = 0$ gives it back $\Rightarrow$ C any real number, including 0

↓

6. CONDITION
$1 = -2 + Ce^0 \Rightarrow C = 3$: $\quad y = -2 + 3e^{2x^2}$

↓

7. CHECK
$y' = 12xe^{2x^2}$ $\qquad 4x(y+2) = 4x \cdot 3e^{2x^2} = 12xe^{2x^2}$ ✓

Worked: $y' = -\frac{x}{y}$ *through* $(3,4)$ *(derives the section 8 circle)*

1. PIECES
$g = -x$ $\quad p = \frac{1}{y}$ $\quad (3,4)$

↓

2. TEMPLATE
$$\int \frac{dy}{\boxed{p(y)}} = \int \boxed{g(x)}\,dx + c$$

↓

3. PLACE
$$\int \frac{dy}{1/y} = \int y\,dy = \int (-x)\,dx \quad \textit{(dividing by } \tfrac{1}{y} \textit{ is multiplying by } y\textit{)}$$

↓

4. INTEGRATE, ×2 EVERY TERM
$$\frac{y^2}{2} = -\frac{x^2}{2} + c \quad \Rightarrow \quad x^2 + y^2 = C \quad (C = 2c)$$

↓

5. CONDITION
$3^2 + 4^2 = C \Rightarrow x^2 + y^2 = 25$

↓

6. CHECK
implicit differentiation gives $y' = -\frac{x}{y}$ (section 8) ✓

Worked: (square-root ending) $y' = \frac{x}{y}$, $y(0) = -2$

1. PIECES
$g = x$ $\quad p = \frac{1}{y}$ $\quad (0,-2)$

↓

2. TEMPLATE
$$\int \frac{dy}{\boxed{p(y)}} = \int \boxed{g(x)}\,dx + c$$

↓

3. PLACE, INTEGRATE
$$\int y\,dy = \int x\,dx \quad \Rightarrow \quad \frac{y^2}{2} = \frac{x^2}{2} + c \quad \Rightarrow \quad y^2 = x^2 + C$$

↓

4. FREE y
$y = \pm\sqrt{x^2 + C}$ *(the square root brings ±)*

↓

5. CONDITION PICKS THE SIGN
$-2 = \pm\sqrt{C}$. A root is never negative, so keep $-$: $\quad \sqrt{C} = 2$, $\quad C = 4$ $\Rightarrow$ $y = -\sqrt{x^2+4}$

↓

6. CHECK
$y' = \frac{-x}{\sqrt{x^2+4}}$ $\qquad \frac{x}{y} = \frac{x}{-\sqrt{x^2+4}}$, the same ✓

Worked: (exponential ending) $y' = 6x^2e^{3y}$, $y(0) = 0$

1. PIECES
$g = 6x^2$ $\quad p = e^{3y}$ $\quad (0,0)$

↓

2. TEMPLATE
$$\int \frac{dy}{\boxed{p(y)}} = \int \boxed{g(x)}\,dx + c$$

↓

3. PLACE
$$\int \frac{dy}{e^{3y}} = \int e^{-3y}\,dy = \int 6x^2\,dx \quad \textit{(divide: the exponent flips sign)}$$

↓

4. INTEGRATE
$\frac{e^{-3y}}{-3} = 2x^3 + c$ *(the same -3 in both places)*

↓

5. FREE y
$\times(-3)$: $e^{-3y} = -6x^3 + C$ $\quad$ ln: $-3y = \ln(C - 6x^3)$ $\quad \div(-3)$: $y = -\frac{1}{3}\ln(C - 6x^3)$

↓

6. CONDITION
$0 = -\frac{1}{3}\ln C \Rightarrow C = 1$: $\quad y = -\frac{1}{3}\ln(1 - 6x^3)$

↓

7. INTERVAL (SECTION 8)
ln needs $1 - 6x^3 > 0$: $\quad x < 6^{-1/3} \approx 0.550 \Rightarrow (-\infty, 0.550)$

↓

8. CHECK
$y' = \frac{6x^2}{1 - 6x^3}$ $\qquad e^{3y} = \frac{1}{1 - 6x^3}$, so $6x^2e^{3y} = y'$ ✓

Exponentials in any base

$a^t = e^{(\ln a)t}$, so $x = a^t$ solves $x' = kx$ with $k = \ln a$, a constant. The section 9 model $P = P_0e^{at}$ is the separable solution of $P' = aP$.

Quick example	Work → answer
$x = 2^t$	$k = \ln 2 = 0.693$: growth
$x = (0.5)^t$	$k = \ln 0.5 = -0.693$: decay $(0 < a < 1)$

14. autonomous equations

THE IDEA
$y' = f(y)$ has no x on the right. The slope depends only on the height. So every horizontal row of the field has one slope, and a solution slid left or right is still a solution.

Formula	In words
$y' = f(y)$	separable with $g = 1$: $\int \frac{dy}{f(y)} = x + c$
$f(k) = 0$	equilibria = lost solutions of the separation
$y'' = f'(y)\,f(y)$	concavity, by the chain rule

Sign of $f'(y)\,f(y)$	Curve
> 0	concave up
< 0	concave down
$f'(y) = 0$ between equilibria	inflection point

Worked: where do the curves of $y' = y(2-y)$ bend?

1. PIECES
$f = y(2-y)$, zero at 0, 2 $\quad$ $f' = 2 - 2y$, zero at $y = 1$

↓

2. TEMPLATE
sign of $\boxed{f'} \cdot \boxed{f}$, one test per strip

↓

3. PLACE
$0 < y < 1$: $(+)(+)$ up $\quad$ $1 < y < 2$: $(-)(+)$ down $\quad$ $y > 2$: $(-)(-)$ up $\quad$ $y < 0$: $(+)(-)$ down

↓

4. READ
curves rising from 0 to 2 bend up, flip at $y = 1$, then flatten into 2: the **S-curve** (picture in 12)

↓

5. CHECK
$y = \frac{2}{1+e^{-2x}}$ (next example) is 1 at $x = 0$, its steepest point ✓

Worked: solve $y' = y(2-y)$, $\quad y(0) = 1$

1. PIECES
$f = y(2-y)$ $\quad$ $(0,1)$

↓

2. TEMPLATE
$$\int \frac{dy}{\boxed{f(y)}} = x + c$$

↓

3. PLACE, PARTIAL FRACTIONS
$$\int \frac{dy}{y(2-y)} = x + c \qquad \frac{1}{y(2-y)} = \frac{1}{2}\left(\frac{1}{y} + \frac{1}{2-y}\right)$$

↓

4. INTEGRATE
$\frac{1}{2}\left(\ln|y| - \ln|2-y|\right) = x + c \;\Rightarrow\; \ln\left|\frac{y}{2-y}\right| = 2x + 2c$

↓

5. FREE y, CONDITION
$\frac{y}{2-y} = Ce^{2x}$ $\quad$ $\frac{1}{2-1} = C$, so $C = 1$ $\quad$ $y = (2-y)e^{2x} \Rightarrow y = \frac{2}{1+e^{-2x}}$

↓

6. LOST SOLUTIONS
$y = 0$: $C = 0$ gives it $\quad$ $y = 2$: no C gives it, list it separately

↓

7. CHECK
$x = 0$: $\frac{2}{2} = 1$ ✓ $\quad$ $x \to \infty$: $y \to 2$, the stable equilibrium ✓

15. exact equations

THE IDEA
Think of a hill $F(x,y)$. A solution curve walks along one height, $F = c$: a level curve. If M is the x slope of the hill and N is the y slope, rebuild the hill from its slopes.

$$\boxed{M(x,y)}\,dx + \boxed{N(x,y)}\,dy = 0 \qquad F_x = \boxed{M},\;\; F_y = \boxed{N}$$

Formula	In words
$M\,dx + N\,dy = 0$	M multiplies dx, N multiplies dy, with signs
$M + Ny' = 0$	the same equation, divided by dx
$\frac{d}{dx}F(x,y) = M + Ny'$	multivariable chain rule: so $F = c$
$M_y = N_x$	**exactness test** (Clairaut): a cross, M by y, N by x
$F = \int M\,dx + g(y)$	integrate in x. The "constant" is a function of y
$F_y = N$	solves for $g'(y)$. Only y may remain

Procedure

1. Rearrange to $M\,dx + N\,dy = 0$. A term that crosses $=$ flips sign.
2. Test $M_y = N_x$. If they differ, the equation is not exact: go to 17.
3. $F = \int M\,dx + g(y)$.
4. $F_y = N$: differentiate every term, $g(y) \to g'(y)$ too. Solve for $g'(y)$.
5. Integrate $g'(y)$ to get $g(y)$. No constant needed.
6. Answer: $F(x,y) = c$, one equation. IVP: put in (x_0, y_0) for c.

Rearranging: $y' = \frac{A}{B} \to B\,dy = A\,dx \to A\,dx - B\,dy = 0$. Put the minus into N.

Quick example	Work → answer
$(y + \cos x)\,dx + (x + 2y)\,dy$	$M_y = 1$, $N_x = 1$: exact
$(-2xy - 1)\,dx + x^2\,dy$	$M_y = -2x$, $N_x = 2x$: not exact *(17)*
$(kxy^2 + 1)\,dx + 3x^2y\,dy$: find k	$M_y = 2kxy$, $N_x = 6xy$: $\quad 2k = 6, \quad k = 3$

Worked: $(2xy + 5)\,dx + (x^2 + 6y)\,dy = 0$, through $(1,1)$

1. PIECES (TEST FIRST)
$M = 2xy + 5$ $\quad$ $N = x^2 + 6y$ $\quad$ $(1,1)$
$M_y = 2x$, $\quad N_x = 2x$ $\quad$ equal ⇒ exact

↓

2. TEMPLATE
$$F = \int \boxed{M}\,dx + g(y), \quad \text{then } F_y = \boxed{N}$$

↓

3. PLACE
$$F = \int (2xy + 5)\,dx = x^2y + 5x + g(y)$$
$F_y = x^2 + g'(y) = x^2 + 6y \Rightarrow g' = 6y, \quad g = 3y^2$

↓

4. ANSWER, CONDITION
$x^2y + 5x + 3y^2 = c$ $\quad$ $(1,1)$: $1 + 5 + 3 = 9 \Rightarrow x^2y + 5x + 3y^2 = 9$

↓

5. CHECK (IMPLICIT DIFFERENTIATION)
$2xy + x^2y' + 5 + 6yy' = 0 \Rightarrow (2xy + 5) + (x^2 + 6y)y' = 0$ ✓

Worked: (rearrange first) $y' = \dfrac{2x + ye^x}{3y^2 - e^x}$

1. PIECES (REARRANGE FIRST)
$(3y^2 - e^x)\,dy = (2x + ye^x)\,dx \Rightarrow (2x + ye^x)\,dx - (3y^2 - e^x)\,dy = 0$
$M = 2x + ye^x$ $\quad$ $N = e^x - 3y^2$ *(minus moved in)* $\quad$ $M_y = e^x = N_x$: exact

↓

2. TEMPLATE
$$F = \int \boxed{M}\,dx + g(y), \quad \text{then } F_y = \boxed{N}$$

↓

3. PLACE
$$F = \int (2x + ye^x)\,dx = x^2 + ye^x + g(y)$$
$F_y = e^x + g'(y) = e^x - 3y^2 \Rightarrow g' = -3y^2, \quad g = -y^3$

↓

4. ANSWER
$x^2 + ye^x - y^3 = c$

↓

5. CHECK
$2x + ye^x + e^xy' - 3y^2y' = 0 \Rightarrow y' = \dfrac{2x + ye^x}{3y^2 - e^x}$ ✓

Traps

Trap: Forgetting $+g(y)$ in step 3.
Trap: Writing $g(y)$ instead of $g'(y)$ after you differentiate.
Trap: Adding $\int N\,dy$ on top of F: shared terms count twice.
Trap: Swapping M and N. M is always the dx coefficient.

16. first order linear equations

THE IDEA
Multiply the equation by a factor μ chosen so that the left side becomes one product-rule derivative, $(\mu y)'$. Then one integration finishes the job.

$$y' + \boxed{p(x)}\,y = \boxed{q(x)} \qquad \mu = e^{\int \boxed{p}\,dx} \qquad \mu y = \int \mu\,\boxed{q}\,dx + c$$

Formula	In words
$y' + p(x)\,y = q(x)$	**standard form**: y' alone, coefficient 1 *(Trench writes $f(x)$ for q)*
$\mu = e^{\int p\,dx}$	integrating factor. No $+c$
$(\mu y)' = \mu q$	the left side after multiplying
$y = \dfrac{1}{\mu}\left(\int \mu q\,dx + c\right)$	divide by μ, c too

Why: $(\mu y)' = \mu y' + \mu' y$. This equals $\mu y' + p\mu y$ when $\mu' = p\mu$. That is a separable equation, and its solution is $\mu = e^{\int p}$.
Trench's way (variation of parameters): $y = uy_1$, with $y_1 = e^{-\int p}$ solving $y' + py = 0$. Then $u' = \frac{q}{y_1} = \mu q$: the same integral.

1. Standard form: divide by the coefficient of y'.
2. Read p (with its sign) and q.
3. $\mu = e^{\int p\,dx}$. Simplify $e^{\ln}$.
4. $\mu y = \int \mu q\,dx + c$.
5. Divide by μ. Interval: where p and q are continuous, through x_0.

Worked: $y' + 3y = e^{2x}$

1. PIECES
$p = 3 \qquad q = e^{2x}$

↓

2. TEMPLATE
$\mu = e^{\int \boxed{p}\,dx} \qquad \mu y = \int \mu\,\boxed{q}\,dx + c$

↓

3. PLACE
$\mu = e^{\int 3\,dx} = e^{3x} \qquad e^{3x}y = \int e^{3x}e^{2x}\,dx = \dfrac{e^{5x}}{5} + c$

↓

4. DIVIDE BY μ
$y = \dfrac{e^{2x}}{5} + ce^{-3x}$ *(c divided too)*

↓

5. CHECK
$y' + 3y = \frac{2e^{2x}}{5} - 3ce^{-3x} + \frac{3e^{2x}}{5} + 3ce^{-3x} = e^{2x}$ ✓

Worked: (standard form first) $xy' + 2y = x^2, \quad y(1) = 1$

1. PIECES (STANDARD FORM: $\div x$)
$y' + \dfrac{2}{x}y = x \qquad p = \frac{2}{x} \qquad q = x \qquad (1,1)$

↓

2. TEMPLATE
$\mu = e^{\int \boxed{p}\,dx} \qquad \mu y = \int \mu\,\boxed{q}\,dx + c$

↓

3. PLACE IN μ
$\mu = e^{\int 2/x\,dx} = e^{2\ln|x|} = x^2$ *($e^{\ln\square} = \square$)*

↓

4. PLACE IN μy
$x^2 y = \int x^2 \cdot x\,dx = \dfrac{x^4}{4} + c \;\Rightarrow\; y = \dfrac{x^2}{4} + \dfrac{c}{x^2}$

↓

5. CONDITION, INTERVAL
$1 = \frac{1}{4} + c \Rightarrow c = \frac{3}{4}$: $y = \dfrac{x^2}{4} + \dfrac{3}{4x^2}$ on $(0,\infty)$ *(p breaks at 0)*

↓

6. CHECK
$xy' = \frac{x^2}{2} - \frac{3}{2x^2}, \quad 2y = \frac{x^2}{2} + \frac{3}{2x^2}, \quad \text{sum} = x^2$ ✓

Quick example	Work → answer
$y' - 2y = 4x$	$p = -2$: $\mu = e^{-2x}$. $\int 4xe^{-2x}dx$ by parts. $y = -2x - 1 + ce^{2x}$
$y' + 3y = 6$	linear *and* separable: $y = 2 + ce^{-3x}$
cooling $T' + kT = kT_m$	$\mu = e^{kt}$: $T = T_m + ce^{-kt}$ *(section 9)*

Trap: Standard form first: in $xy' + 2y = x^2$, p is $\frac{2}{x}$, not 2.
Trap: The sign rides with p: $y' - 2y$ has $p = -2$.
Trap: Divide c by μ too: $\frac{c}{x^2}$, not c.

17. integrating factors: not exact → exact

THE IDEA
If $M_y \neq N_x$, multiply the whole equation by a factor μ that forces the cross test to pass. The mismatch $M_y - N_x$ tells you what μ is.

Formula	In words
$\dfrac{\boxed{M_y} - \boxed{N_x}}{\boxed{N}}$ has only x	$\mu(x) = e^{\int (\text{that})\,dx}$
$\dfrac{\boxed{N_x} - \boxed{M_y}}{\boxed{M}}$ has only y	$\mu(y) = e^{\int (\text{that})\,dy}$ *(flipped top, M below)*
neither	try another method: linear (16), Bernoulli (19)

Why: for $(\mu M)_y = (\mu N)_x$ with $\mu(x)$: $\mu M_y = \mu' N + \mu N_x$, so $\dfrac{\mu'}{\mu} = \dfrac{M_y - N_x}{N}$. That must have no y.
Trap: μ can add or remove a solution where $\mu = 0$ or is undefined. Test those lines in the original equation.

Worked: $(-2xy - 1)\,dx + x^2\,dy = 0$

1. PIECES
$M = -2xy - 1, \quad M_y = -2x \qquad N = x^2, \quad N_x = 2x \qquad$ not equal: not exact

↓

2. TEMPLATE ($\mu(x)$ TEST)
$\dfrac{\boxed{M_y} - \boxed{N_x}}{\boxed{N}}$ must have only $x \qquad \mu = e^{\int (\text{that})\,dx}$

↓

3. PLACE
$\dfrac{-2x - 2x}{x^2} = -\dfrac{4}{x}$: only x ✓ $\qquad \mu = e^{-4\ln|x|} = x^{-4}$

↓

4. MULTIPLY, RETEST
$\left(-2x^{-3}y - x^{-4}\right)dx + x^{-2}\,dy = 0 \qquad M_y = -2x^{-3} = N_x$ ✓

↓

5. SOLVE THE EXACT EQUATION (SECTION 15)
$F = \int \left(-2x^{-3}y - x^{-4}\right)dx = x^{-2}y + \frac{1}{3}x^{-3} + g(y)$
$F_y = x^{-2} + g' = x^{-2} \Rightarrow g = 0 \qquad \dfrac{y}{x^2} + \dfrac{1}{3x^3} = c$, on $x > 0$ or $x < 0$

↓

6. CHECK (SOLVE IT AS LINEAR)
$y' - \frac{2}{x}y = \frac{1}{x^2}$, $\mu = x^{-2}$, $\frac{y}{x^2} = -\frac{1}{3x^3} + c$, the same ✓

Worked: $(\mu(y))\; y\,dx + (2x+y^2)\,dy = 0$

1. PIECES
$M = y$, $M_y = 1$ $\quad$ $N = 2x+y^2$, $N_x = 2$

↓

2. TEMPLATES (BOTH TESTS)
$\frac{M_y - N_x}{N}$ only x? $\quad$ $\frac{N_x - M_y}{M}$ only y?

↓

3. PLACE
$\frac{1-2}{2x+y^2}$ has a y no $\quad$ $\frac{2-1}{y} = \frac{1}{y}$: only y ✓ $\quad$ $\mu = e^{\ln|y|} = y$

↓

4. MULTIPLY, RETEST
$y^2\,dx + (2xy+y^3)\,dy = 0$ $\quad$ $M_y = 2y = N_x$ ✓

↓

5. SOLVE THE EXACT EQUATION (SECTION 15)
$F = xy^2 + g(y)$, $\quad F_y = 2xy + g' = 2xy + y^3 \Rightarrow g = \frac{y^4}{4}$ $\quad$ $xy^2 + \frac{y^4}{4} = c$

↓

6. CHECK
$\frac{d}{dx}$: $y^2 + (2xy+y^3)y' = 0$, then $\div y$ gives the original ✓
$\mu = 0$ at $y = 0$: in the original, $0\,dx + 2x\cdot 0 = 0$, so $y = 0$ is a solution too

18. transformations (substitutions)

THE IDEA
Rename part of the equation with a new letter. If you pick the right part, the equation in the new letter is one you can already solve (separable or linear). Solve it, then put the old letters back.

Formula	In words
$y' = q\left(\frac{y}{x}\right)$	right side depends only on $\frac{y}{x}$: **homogeneous** (this meaning only here)
$y = ux$, $\quad y' = u + xu'$	product rule on ux
$xu' = q(u) - u$	separable in u and x
$q(u_0) = u_0$	constant u: lost solution $y = u_0 x$
$y' = f(ax+by+c)$	right side depends only on one combination
$v = ax+by+c$, $\quad v' = a + bf(v)$	then $v' = a + bf(v)$: autonomous in v

Test for $\frac{y}{x}$: put $y = ux$. If every x cancels, it is homogeneous. Quick version: every term on top and bottom has the same total degree.
Trap: Two meanings of homogeneous. Section 16: $q = 0$. Here: $y' = q(y/x)$.
Trap: Substitute y' too. $y = ux$ means $y' = u + xu'$, not u'.

Quick example	Work → answer
$y' = \frac{y-x}{x+y}$	$\div x$: $\frac{u-1}{1+u}$ homogeneous
$y' = \frac{x^2+y}{x}$	$x+u$: an x is left not y/x *(linear: 16)*
$y' = \sin(x-y)$	$v = x-y$: $\quad v' = 1 - \sin v$ separable

Worked: $y' = \frac{x^2+y^2}{xy}$, $\quad y(1) = 2$

1. PIECES ($\div x^2$ TOP AND BOTTOM)
$\frac{1+(y/x)^2}{y/x} \;\Rightarrow\; q(u) = \frac{1}{u} + u$ $\quad$ $(1,2)$

↓

2. TEMPLATE
$u + xu' = \boxed{q(u)}$

↓

3. PLACE, SEPARATE
$u + xu' = \frac{1}{u} + u \Rightarrow xu' = \frac{1}{u} \Rightarrow u\,du = \frac{dx}{x}$

↓

4. INTEGRATE, BACK TO y
$\frac{u^2}{2} = \ln|x| + c \Rightarrow u^2 = 2\ln|x| + C \Rightarrow y^2 = x^2(2\ln|x| + C)$

↓

5. CONDITION
$2^2 = 1^2(0+C) \Rightarrow C = 4$. $\quad y > 0$: $\quad y = x\sqrt{2\ln x + 4}$ on (e^{-2}, ∞)

↓

6. CHECK
$S = 2\ln x + 4$: $\quad y' = \sqrt{S} + \frac{1}{\sqrt{S}}$ $\quad$ right side $\frac{x^2 + x^2 S}{x^2\sqrt{S}} = \frac{1+S}{\sqrt{S}}$, the same ✓

Worked: $y' = (x+y+1)^2$, $\quad y(0) = -1$

1. PIECES
one combination: $v = x+y+1$, so $a = 1$, $b = 1$ $\quad$ $f(v) = v^2$ $\quad$ $(0,-1)$, so $v(0) = 0$

↓

2. TEMPLATE
$v' = a + b\cdot\boxed{f(v)}$

↓

3. PLACE, SEPARATE
$v' = 1 + v^2$ *(autonomous)* $\Rightarrow$ $\frac{dv}{1+v^2} = dx$

↓

4. INTEGRATE, BACK TO y
$\arctan v = x + c \Rightarrow v = \tan(x+c) \Rightarrow y = \tan(x+c) - x - 1$

↓

5. CONDITION
$-1 = \tan c - 1 \Rightarrow c = 0$: $\quad y = \tan x - x - 1$

↓

6. CHECK
$y' = \sec^2 x - 1 = \tan^2 x$, $\quad$ and $(x+y+1)^2 = \tan^2 x$ ✓

19. Bernoulli equations

THE IDEA
A Bernoulli equation is a linear equation with one extra power of y on the right. The substitution $v = y^{1-n}$ removes the power. The equation in v is linear, so section 16 finishes it.

$$y' + \boxed{p(x)}y = \boxed{q(x)}y^{\boxed{n}} \quad\Rightarrow\quad v' + (1-\boxed{n})\,\boxed{p}\,v = (1-\boxed{n})\,\boxed{q}$$

Formula	In words
$n \neq 0, 1$	$n = 0$ or 1 is already linear
$v = y^{1-n}$	the new letter
$v' = (1-n)y^{-n}y'$	chain rule
divide the DE by y^n	$y^{-n}y' + py^{1-n} = q$, then put in v
$y = v^{1/(1-n)}$	back to y at the end
$y = 0$	a lost solution if $n > 0$

Trench's way: $y = uy_1$ with $y_1 = e^{-\int p}$ gives a separable equation in u. Both ways give the same y.
Trap: The factor $(1-n)$ multiplies both p and q.
Trap: Undo v at the end. If $v^{1/(1-n)}$ is an even root ($n = 3$: $v^{-1/2}$, $\quad n = -1$: $v^{1/2}$), write $\pm$. The condition picks the sign.

Quick example	Work → answer
$y' - y = e^x y^{1/2}$	$n = \frac{1}{2}$: $v = y^{1/2}$
$y' = y - y^3$	$n = 3$, $p = -1$, $q = -1$ *(also autonomous)*
$y' + \frac{y}{x} = \frac{1}{y}$	$n = -1$: $v = y^2$

Worked: $y' + \frac{1}{x}y = xy^2$, $x > 0$

1. PIECES
$p = \frac{1}{x}$ $q = x$ $n = 2$ $\Rightarrow$ $1-n = -1$, $v = y^{-1}$

↓

2. TEMPLATE
$v' + (1-\boxed{n})\boxed{p}v = (1-\boxed{n})\boxed{q}$

↓

3. PLACE
$v' + (1-2)\frac{1}{x}v = (1-2)x \;\Rightarrow\; v' - \frac{1}{x}v = -x$

↓

4. SOLVE THE LINEAR EQUATION (SECTION 16)
$\mu = e^{-\ln x} = \frac{1}{x}$ $\left(\frac{v}{x}\right)' = -1 \Rightarrow \frac{v}{x} = -x + c \Rightarrow v = cx - x^2$

↓

5. BACK TO y
$y = v^{-1} = \dfrac{1}{cx - x^2}$, and $y = 0$ *(lost when we divided by y^2)*

↓

6. CHECK
$D = cx - x^2$: $y' = \dfrac{2x-c}{D^2}$, $\dfrac{y}{x} = \dfrac{c-x}{D^2}$, sum $\dfrac{x}{D^2} = xy^2$ ✓

Worked: $y' + y = xy^3$, $y(0) = 1$

1. PIECES
$p = 1$ $q = x$ $n = 3$ $\Rightarrow$ $1-n = -2$, $v = y^{-2}$ $v(0) = 1^{-2} = 1$

↓

2. TEMPLATE
$v' + (1-\boxed{n})\boxed{p}v = (1-\boxed{n})\boxed{q}$

↓

3. PLACE
$v' + (1-3)(1)v = (1-3)x \;\Rightarrow\; v' - 2v = -2x$

↓

4. SOLVE THE LINEAR EQUATION (SECTION 16)
$\mu = e^{-2x}$ $e^{-2x}v = \int -2xe^{-2x}dx = xe^{-2x} + \frac{1}{2}e^{-2x} + c$ *(by parts)*
$v = x + \frac{1}{2} + ce^{2x}$

↓

5. CONDITION
$1 = 0 + \frac{1}{2} + c \Rightarrow c = \frac{1}{2}$: $v = x + \frac{1}{2} + \frac{1}{2}e^{2x}$

↓

6. BACK TO y
$y = \pm v^{-1/2}$. $y(0) = 1 > 0$, so keep $+$: $y = \dfrac{1}{\sqrt{x + \frac{1}{2} + \frac{1}{2}e^{2x}}}$

↓

7. CHECK
$v' - 2v = 1 + e^{2x} - 2x - 1 - e^{2x} = -2x$ ✓ $y(0) = \frac{1}{\sqrt{1}} = 1$ ✓

20. applications: growth and decay

THE IDEA
"The rate of change is proportional to the amount" gives $Q' = aQ$, so the amount is an exponential. Most problems give you two facts (a start value and one reading, a half-life, a doubling time). Use one fact for Q_0 and the other for the rate.

Formula	In words
$Q' = aQ \Rightarrow Q = Q_0 e^{a(t-t_0)}$	(amount) = (start)(growth factor). $a > 0$ grows, $a < 0$ decays
$Q' = -kQ$, $Q = Q_0 e^{-kt}$	radioactive decay. $k > 0$ is the **decay constant**
$\tau = \dfrac{\ln 2}{k}$, $k = \dfrac{\ln 2}{\tau}$	**half-life** $\leftrightarrow$ decay constant
$Q = Q_0\left(\frac{1}{2}\right)^{t/\tau}$	same decay, counted in half-lives
$t = \dfrac{\tau \ln(Q_0/Q)}{\ln 2}$	time to fall from Q_0 to Q (carbon dating)
$Q = Q_0\left(1 + \frac{r}{n}\right)^{nt}$	compounded n times a year
$Q = Q_0 e^{rt}$, $Q_0 = Qe^{-rt}$	compounded continuously, and the deposit needed now (present value)
$Q' = s - kQ \Rightarrow$ $Q = \dfrac{s}{k} + \left(Q_0 - \dfrac{s}{k}\right)e^{-kt}$	made at rate s, decays at kQ. **Steady state** $\frac{s}{k}$ *(Trench writes a for s)*
$Q' = rQ + d \Rightarrow$ $Q = Q_0 e^{rt} + \dfrac{d}{r}\left(e^{rt} - 1\right)$	savings: interest r plus deposits d per year, made continuously

Trap: Write a rate as a decimal: $5\frac{1}{2}\%$ is $r = 0.055$.
Trap: Keep t in the units of τ or r (years with years).
Trap: The half-life does not depend on how much you start with.

Quick example	Work $\rightarrow$ answer
\$500 at 4%, continuous, 8 yr	$500e^{0.32} = \$688.56$
need \$20,000 in 15 yr at 6%, continuous	$Q_0 = 20000e^{-0.9} = \$8131.39$
\$1000 at 6%, monthly, 5 yr	$1000(1.005)^{60} = \$1348.85$
made at 3 g/yr, $k = 0.1$	steady state $\frac{3}{0.1} = 30$ g, from any Q_0

Worked: half-life 1200 yr, 6 g now: amount after 300 yr, and time until 2 g

1. PIECES
$Q_0 = 6$ $\tau = 1200$ time $t = 300$ target $Q = 2$

↓

2. TEMPLATES
$Q = \boxed{Q_0}\left(\frac{1}{2}\right)^{t/\boxed{\tau}}$ $t = \dfrac{\boxed{\tau}\ln(\boxed{Q_0}/Q)}{\ln 2}$

↓

3. PLACE
$Q(300) = 6\left(\frac{1}{2}\right)^{300/1200} = 6(0.8409) = 5.05$ g
$t = \dfrac{1200\ln(6/2)}{\ln 2} = 1200(1.585) = 1902$ yr

↓

4. CHECK
1902 yr is 1.585 half-lives. After 1 half-life $6 \rightarrow 3$, and 2 comes a bit later ✓

Worked: (reverse: carbon dating) bones hold 30% of the live amount of carbon-14, $\tau = 5570$ yr

1. PIECES
$Q_0/Q = \frac{1}{0.30}$ *(only the ratio matters)* $\tau = 5570$

↓

2. TEMPLATE
$t = \dfrac{\boxed{\tau}\ln(\boxed{Q_0/Q})}{\ln 2}$

↓

3. PLACE
$t = \dfrac{5570\ln(1/0.30)}{0.6931} = \dfrac{5570(1.2040)}{0.6931} = 9675$ yr

↓

4. CHECK
$\left(\frac{1}{2}\right)^{9675/5570} = \left(\frac{1}{2}\right)^{1.737} = 0.30$ ✓ *(use the half-life the problem gives)*

Worked: savings: $2000 ***now,*** $3000 ***per year added continuously,*** 5% ***continuous. Find*** $Q(10)$

1. PIECES
rate in = interest + deposits: $Q' = 0.05Q + 3000 \qquad Q_0 = 2000$

↓

2. TEMPLATE (LINEAR, SECTION 16)
$$Q = \boxed{Q_0}e^{\boxed{r}t} + \frac{\boxed{d}}{\boxed{r}}\left(e^{\boxed{r}t} - 1\right)$$

↓

3. PLACE, $t = 10$
$$Q = 2000e^{0.5} + \frac{3000}{0.05}\left(e^{0.5} - 1\right) = 3297.44 + 38923.28 = \$42{,}220.72$$

↓

4. CHECK
money put in: $2000 + 30{,}000 = \$32{,}000$. The rest is interest ✓

21. applications: cooling and mixing

THE IDEA
Both are "rate in minus rate out." Cooling: the gap to the room shrinks exponentially. Mixing: salt flows in at a fixed rate and leaves in proportion to how much is in the tank. Both give a linear equation (section 16).

Formula	In words
$T' = -k(T - T_m)$	Newton: the rate is proportional to the gap from the room T_m
$T = \boxed{T_m} + (\boxed{T_0} - \boxed{T_m})e^{-\boxed{k}t}$	(temp) = (room) + (start gap)(fade)
$e^{-kt_1} = \dfrac{T_1 - T_m}{T_0 - T_m}$	one reading T_1 at t_1 gives k
$t = -\dfrac{1}{k}\ln\dfrac{T - T_m}{T_0 - T_m}$	time to reach a temperature T
$Q' = (\text{rate in}) - (\text{rate out})$	Q is the amount of salt, not the concentration
$\text{rate in} = c_{in}\, r_{in}$	(lb/gal)(gal/min) = lb/min
$\text{rate out} = \dfrac{Q}{V(t)}\, r_{out}$	(concentration in the tank)(flow out)
$V(t) = V_0 + (r_{in} - r_{out})\,t$	volume. The tank overflows when V reaches capacity

Shortcut: with one reading T_1 at t_1, the fade factor is a ratio to a power. No k needed:

$$T = \boxed{T_m} + (\boxed{T_0} - \boxed{T_m})\left(\frac{\boxed{T_1} - \boxed{T_m}}{\boxed{T_0} - \boxed{T_m}}\right)^{t/\boxed{t_1}}$$

Trap: The tank concentration is $\frac{Q}{V}$, not c_{in}. Salt leaves at the tank's concentration.
Trap: If $r_{in} \neq r_{out}$, V changes, so the rate out has t in it.
Trap: Equal flows: $Q \to c_{in}V$, the incoming concentration times the volume.

Worked: cooling: room $20°$***, tea*** $95°$***,*** $70°$ ***after*** 5 ***min. Find*** $T(15)$***. When is it*** $40°$***?***

1. PIECES
$T_m = 20 \qquad T_0 = 95 \qquad$ reading $T_1 = 70$ at $t_1 = 5$

↓

2. TEMPLATE (SHORTCUT)
$$T = \boxed{T_m} + (\boxed{T_0} - \boxed{T_m})\left(\frac{\boxed{T_1} - \boxed{T_m}}{\boxed{T_0} - \boxed{T_m}}\right)^{t/\boxed{t_1}}$$

↓

3. PLACE
$$T = 20 + (95 - 20)\left(\frac{70 - 20}{95 - 20}\right)^{t/5} = 20 + 75\left(\tfrac{2}{3}\right)^{t/5} \quad (k = \tfrac{\ln 1.5}{5} = 0.0811)$$

↓

4. EVALUATE $t = 15$
$$T = 20 + 75\left(\tfrac{2}{3}\right)^{3} = 20 + 22.2 = 42.2°$$

↓

5. REVERSE: $T = 40$
$$\left(\tfrac{2}{3}\right)^{t/5} = \tfrac{20}{75} \Rightarrow t = 5 \cdot \frac{\ln(0.2667)}{\ln(2/3)} = 16.3 \text{ min}$$

↓

6. CHECK
$T(15) = 42.2°$ is just above $40°$, so $40°$ comes just after 15 min ✓

Worked: mixing, equal flows: 200 ***L with*** 10 ***g salt. In:*** 0.3 ***g/L at*** 5 ***L/min. Out:*** 5 ***L/min***

1. PIECES
$\text{rate in} = 0.3(5) = 1.5$ g/min $\qquad \text{rate out} = \frac{Q}{200}(5) = \frac{Q}{40} \qquad Q(0) = 10$

↓

2. TEMPLATE
$Q' = \boxed{\text{rate in}} - \boxed{\text{rate out}}$

↓

3. PLACE
$$Q' = 1.5 - \frac{Q}{40} \quad \Rightarrow \quad Q' + \frac{Q}{40} = 1.5$$

↓

4. SOLVE THE LINEAR EQUATION (SECTION 16)
$\mu = e^{t/40} \qquad e^{t/40}Q = 60e^{t/40} + c \qquad Q = 60 + ce^{-t/40}$

↓

5. CONDITION
$10 = 60 + c \Rightarrow c = -50$: $\quad Q = 60 - 50e^{-t/40}$

↓

6. CHECK
$t \to \infty$: $Q \to 60 = 0.3(200)$, the incoming concentration times the volume ✓

Worked: mixing, unequal flows: 100 ***gal with*** 5 ***lb. In:*** 0.5 ***lb/gal at*** 3 ***gal/min. Out:*** 1 ***gal/min. Holds*** 300 ***gal***

1. PIECES
$V = 100 + 2t$ *(gains 2 gal/min, full at* $t = 100$*)* $\qquad \text{rate in} = 0.5(3) = 1.5 \qquad \text{rate out} = \frac{Q}{100+2t}(1) \qquad Q(0) = 5$

↓

2. TEMPLATE
$Q' = \boxed{\text{rate in}} - \boxed{\text{rate out}}$

↓

3. PLACE
$$Q' = 1.5 - \frac{Q}{100 + 2t} \quad \Rightarrow \quad Q' + \frac{1}{100 + 2t}Q = 1.5$$

↓

4. SOLVE THE LINEAR EQUATION (SECTION 16)
$\mu = e^{\frac{1}{2}\ln(100+2t)} = (100 + 2t)^{1/2}$
$$\mu Q = \int 1.5(100 + 2t)^{1/2}dt = 0.5(100 + 2t)^{3/2} + c \quad \Rightarrow \quad Q = 50 + t + \frac{c}{\sqrt{100 + 2t}}$$

↓

5. CONDITION
$5 = 50 + \frac{c}{10} \Rightarrow c = -450$: $\quad Q = 50 + t - \dfrac{450}{\sqrt{100 + 2t}}$

↓

6. AT OVERFLOW, $t = 100$
$Q = 150 - \frac{450}{17.32} = 124.0$ lb. Concentration $\frac{124.0}{300} = 0.413$ lb/gal

↓

7. CHECK
$Q(0) = 50 - 45 = 5$ ✓ $\qquad 0.413 < 0.5$, the incoming concentration ✓

22. applications: elementary mechanics

THE IDEA
Newton: $F = ma$, and $a = v'$. If the force depends only on t and v, then $mv' = F$ is a first order equation for the velocity. Solve for v, then integrate v to get the position y. Up is positive, so gravity is $-mg$.

System	Length	Mass	Force, g
cgs	cm	g	dyne, 980 cm/s^2
mks	m	kg	newton, 9.8 m/s^2
British	ft	slug	pound, 32 ft/s^2

Weight is a force: a 960 lb object has $m = \frac{960}{32} = 30$ slugs.

Formula	In words
$F = ma = mv', \quad v = y'$	force, velocity, position
$mv' = -mg - kv$	linear drag: resistance kv always opposes motion
$v = -\frac{mg}{k} + \left(v_0 + \frac{mg}{k}\right)e^{-kt/m}$	solution (linear, section 16)
$v_{term} = -\frac{mg}{k}$	**terminal velocity**: v as $t \to \infty$
$mv' = -mg + kv^2$	square drag, falling ($v \le 0$): drag is up
$mv' = -mg - kv^2$	square drag, rising ($v \ge 0$): drag is down
$v_{term} = -\sqrt{mg/k}$	square drag terminal velocity
$T_{top} = \sqrt{\frac{m}{gk}}\arctan\left(v_0\sqrt{\frac{k}{mg}}\right)$	square drag: time to the top

Find k: "3 lb for each ft/s" means $k = 3$. "8 N at 2 m/s" with square drag means $k(2)^2 = 8$.

Escape velocity. Gravity weakens with distance: $y'' = -\frac{gR^2}{(y+R)^2}$. Chain rule: $y'' = \frac{dv}{dt} = v\frac{dv}{dy}$, which makes it separable in v and y.

Formula	In words
$v\frac{dv}{dy} = -\frac{gR^2}{(y+R)^2}$	R is Earth's radius, y is altitude, burnout at $y = h$
$v^2 = v_0^2 + 2gR^2\left(\frac{1}{y+R} - \frac{1}{h+R}\right)$	speed at altitude y
$v_e = \sqrt{\frac{2gR^2}{h+R}}$	**escape velocity**: if $v_0 \ge v_e$, it never falls back
$\frac{gR^2}{y_m+R} = \frac{gR^2}{h+R} - \frac{v_0^2}{2}$	maximum altitude y_m if $v_0 < v_e$ (set $v = 0$)

Trap: Pick a sign for "up" and keep it. With up positive, a falling object has $v < 0$.

Trap: Square drag: choose $+kv^2$ or $-kv^2$ so the drag opposes the motion.

Quick example	Work → answer
960 lb, drag 3 lb per ft/s	$m = 30, \quad v_{term} = -\frac{960}{3} = -320$ ft/s
2 kg, square drag 0.4 N at 2 m/s	$k = 0.1, \quad v_{term} = -\sqrt{196} = -14$ m/s
same, thrown up at 14 m/s	$T_{top} = \sqrt{\frac{2}{0.98}}\arctan(1) = 1.429 \cdot \frac{\pi}{4} = 1.12$ s
escape from the surface, $h = 0$	$\sqrt{2gR} = \sqrt{2(9.8)(6.37\times10^6)} = 11.2$ km/s

Worked: a 640 lb object dropped from rest, drag 4 lb per ft/s. Find $v(t)$ and $y(t)$

1. PIECES
$m = \frac{640}{32} = 20$ slugs $\qquad k = 4 \qquad v_0 = 0, y(0) = 0 \qquad g = 32$

↓

2. TEMPLATE
$v = -\frac{mg}{k} + \left(v_0 + \frac{mg}{k}\right)e^{-kt/m}$

↓

3. PLACE
$\frac{20(32)}{4} = 160, \quad \frac{4}{20} = 0.2: \quad v = -160 + (0+160)e^{-0.2t} = -160\left(1 - e^{-0.2t}\right)$ ft/s

↓

4. POSITION ($y = \int v\,dt$)
$y = -160t - 800e^{-0.2t} + c. \quad y(0) = 0 \Rightarrow c = 800: \quad y = -160t + 800\left(1 - e^{-0.2t}\right)$

↓

5. CHECK
$v(0) = 0$ ✓ $\quad t \to \infty$: $v \to -160$, the terminal velocity ✓ $\quad 20v' = -640 - 4v$ ✓

Worked: (reverse: escape) burnout at $h = R$, one Earth radius up. What v_0 escapes?

1. PIECES
$h = R \qquad g = 9.8, \quad R = 6.37 \times 10^6$ m

↓

2. TEMPLATE
$v_e = \sqrt{\frac{2gR^2}{h+R}}$

↓

3. PLACE
$v_e = \sqrt{\frac{2gR^2}{R+R}} = \sqrt{gR} = \sqrt{9.8(6.37\times10^6)} = 7.90$ km/s

↓

4. CHECK
smaller than 11.2 km/s from the ground: gravity is weaker up there ✓

23. homogeneous linear second order equations

THE IDEA
A second order linear equation needs two independent solutions. Every solution is then $c_1y_1+c_2y_2$. The **Wronskian** W is the test for "independent": if $W\neq 0$, the pair can match any $y(x_0)$ and $y'(x_0)$.

Formula	In words
$y''+p(x)y'+q(x)y=f(x)$	standard form. **Homogeneous** if $f=0$ (the section 16 meaning)
$P_0y''+P_1y'+P_2y=0$	divide by P_0: $p=\frac{P_1}{P_0}$, $q=\frac{P_2}{P_0}$. Skip zeros of P_0
p, q continuous on $(a,b)\ni x_0$	the IVP $y(x_0)=k_0$, $y'(x_0)=k_1$ has exactly one solution there
$y=c_1y_1+c_2y_2$	**superposition**: sums of solutions are solutions
$\frac{y_2}{y_1}\neq$ constant	**linearly independent**: neither is a multiple of the other
$W=\begin{vmatrix}y_1 & y_2\\ y_1' & y_2'\end{vmatrix}=y_1y_2'-y_1'y_2$	**Wronskian**
$W\neq 0$	$\{y_1,y_2\}$ is a **fundamental set**: general solution $c_1y_1+c_2y_2$
$W=Ke^{-\int p\,dx}$	**Abel's formula**: W is never 0, or always 0
$c_1=\frac{k_0y_2'-k_1y_2}{W}$, $c_2=\frac{k_1y_1-k_0y_1'}{W}$	IVP constants (all at x_0), Cramer's rule
$y_2=uy_1$, $u'=\frac{e^{-\int p\,dx}}{y_1^2}$	**reduction of order**: a second solution from one
$p=-\frac{W'}{W}$, $q=\frac{y_1'y_2''-y_1''y_2'}{W}$	build the equation from y_1, y_2

Trap: Standard form first. Abel and reduction of order read p from $y''+py'+qy=0$.
Trap: The interval of a solution must avoid zeros of P_0 and contain x_0 (section 8).
Trap: W of two functions that do not come from one such equation can be 0 at some points only. For solutions, W is never 0 or always 0.

Which way? — solutions and equation

Given → find	Do this
y_1, y_2 and the DE → general solution	verify each (section 4), then $W\neq 0$
y_1, y_2 → the DE	$p=-\frac{W'}{W}$, $q=\frac{y_1'y_2''-y_1''y_2'}{W}$
one solution y_1 → a second	reduction of order
DE only → W up to K	Abel: $W=Ke^{-\int p}$

Quick example	Work → answer
$\{e^{2x}, 3e^{2x}\}$	ratio 3: dependent, $W=0$
$\{\sin 2x, \sin x\cos x\}$	$\sin 2x=2\sin x\cos x$: dependent
$\{\cos 3x, \sin 3x\}$	$W=3\cos^2 3x+3\sin^2 3x=3$: independent
$\{e^x, e^{-x}\}$	$W=e^x(-e^{-x})-e^xe^{-x}=-2$
$\{x, x^2\}$	$W=x(2x)-1\cdot x^2=x^2$: 0 at $x=0$ only. Independent on $(0,\infty)$

Worked: $y''-2y'-3y=0$**:** $y_1=e^{3x}$, $y_2=e^{-x}$, $y(0)=2$, $y'(0)=10$

1. PIECES
$y_1=e^{3x}$, $y_1'=3e^{3x}$ $\quad y_2=e^{-x}$, $y_2'=-e^{-x}$ $\quad p=-2$ $\quad k_0=2, k_1=10$
verify each (section 4): $9e^{3x}-6e^{3x}-3e^{3x}=0$ ✓ $\quad e^{-x}+2e^{-x}-3e^{-x}=0$ ✓

↓

2. TEMPLATES
$W=\boxed{y_1}\boxed{y_2'}-\boxed{y_1'}\boxed{y_2}$ $\quad c_1\boxed{y_1(0)}+c_2\boxed{y_2(0)}=\boxed{k_0}$, $\quad c_1\boxed{y_1'(0)}+c_2\boxed{y_2'(0)}=\boxed{k_1}$

↓

3. PLACE IN W
$W=e^{3x}(-e^{-x})-3e^{3x}e^{-x}=-4e^{2x}\neq 0$: a fundamental set
Abel: $Ke^{-\int(-2)dx}=Ke^{2x}$ ✓ $\quad$ general: $y=c_1e^{3x}+c_2e^{-x}$

↓

4. PLACE THE CONDITIONS, SOLVE
$c_1(1)+c_2(1)=2$ $\quad c_1(3)+c_2(-1)=10$ $\quad$ add: $4c_1=12\Rightarrow c_1=3, c_2=-1$

↓

5. ANSWER, CHECK
$y=3e^{3x}-e^{-x}$ $\quad y(0)=2$ ✓ $\quad y'(0)=9+1=10$ ✓

Worked: $x^2y''-2xy'+2y=0$**:** $y_1=x$, $y_2=x^2$, $y(1)=3$, $y'(1)=4$

1. PIECES (STANDARD FORM: $\div x^2$)
$y''-\frac{2}{x}y'+\frac{2}{x^2}y=0$ $\quad p=-\frac{2}{x}$ $\quad y_1=x, y_1'=1$ $\quad y_2=x^2, y_2'=2x$ $\quad x_0=1$

↓

2. TEMPLATE
$W=\boxed{y_1}\boxed{y_2'}-\boxed{y_1'}\boxed{y_2}$

↓

3. PLACE
$W=x(2x)-1\cdot x^2=x^2$ $\quad$ Abel: $Ke^{-\int(-2/x)dx}=Kx^2$ ✓

↓

4. INTERVAL
$P_0=x^2$ is 0 at $x=0$. $\quad x_0=1\Rightarrow(0,\infty)$, where $W\neq 0$

↓

5. CONDITIONS AT $x_0=1$
$c_1(1)+c_2(1)=3$ $\quad c_1(1)+c_2(2)=4$ $\quad c_2=1, c_1=2$

↓

6. ANSWER, CHECK
$y=2x+x^2$ on $(0,\infty)$ $\quad y(1)=3$ ✓ $\quad y'(1)=2+2=4$ ✓

Worked: (reduction of order) $x^2y''-3xy'+4y=0$, $y_1=x^2$**. *Find*** y_2

1. PIECES (STANDARD FORM: $\div x^2$)
$y''-\frac{3}{x}y'+\frac{4}{x^2}y=0$ $\quad p=-\frac{3}{x}$ $\quad y_1=x^2$

↓

2. TEMPLATE
$u'=\frac{e^{-\int\boxed{p}dx}}{\boxed{y_1}^2}$, $\quad y_2=u\boxed{y_1}$

↓

3. PLACE
$e^{-\int(-3/x)dx}=e^{3\ln x}=x^3$ $\quad u'=\frac{x^3}{(x^2)^2}=\frac{1}{x}$ $\quad u=\ln x$

↓

4. ANSWER
$y_2=\ln x\cdot x^2=x^2\ln x$

↓

5. CHECK
$y_2'=2x\ln x+x$, $\quad y_2''=2\ln x+3$
$x^2(2\ln x+3)-3x(2x\ln x+x)+4x^2\ln x=0$ ✓

Worked: (reverse: build the DE) $y_1=e^x$, $y_2=xe^x$

1. PIECES
$y_1'=y_1''=e^x$ $\quad y_2'=(1+x)e^x$, $y_2''=(2+x)e^x$

↓

2. TEMPLATE
$W=\boxed{y_1}\boxed{y_2'}-\boxed{y_1'}\boxed{y_2}$ $\quad p=-\frac{W'}{W}$ $\quad q=\frac{\boxed{y_1'}\boxed{y_2'}-\boxed{y_1''}\boxed{y_2'}}{W}$

↓

3. PLACE
$W=e^x(1+x)e^x-e^xxe^x=e^{2x}$, $\quad W'=2e^{2x}$
$p=-\frac{2e^{2x}}{e^{2x}}=-2$ $\quad q=\frac{e^x(2+x)e^x-e^x(1+x)e^x}{e^{2x}}=1$

↓

4. ANSWER
$y''-2y'+y=0$

↓

5. CHECK
y_2: $(2+x)e^x-2(1+x)e^x+xe^x=0$ ✓